CLAUDE CODE USAGE ANALYTICS PLATFORM

FROM RAW TELEMETRY TO PREDICTIVE INSIGHTS

CLAUDE CODE USAGE ANALYTICS PLATFORM

- Content
  - End-to-end analytics system for developer AI usage
  - Covers data ingestion → analytics → visualization → ML
  - Built on realistic synthetic telemetry data
- Project Scope
  - 100 engineers
  - 60-day simulation
  - 454,000+ telemetry events
  - Tech Stack
  - Python • SQLite • Pandas • Streamlit • Plotly • scikit-learn

ARCHITECTURE & APPROACH

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Pipeline Overview
Raw JSONL telemetry
→ SQLite ingestion (structured storage)
→ SQL analytics (12 queries)
→ Streamlit dashboard (5 modules)
→ Machine learning models

DATA PROCESSING

- Parses nested telemetry logs into 5 normalized tables
- Joins user metadata (practice, level, location)
- Aggregates cost, usage, sessions, and behavior patterns

DASHBOARD MODULES

- Overview — high-level KPIs
- Model Analysis — cost & token usage
- Developer Behavior — usage patterns & tools
- Team Insights — organizational breakdown
- Predictive Analytics — forecasting & anomaly detection

DEVELOPER BEHAVIOR INSIGHTS

HOW ENGINEERS USE CLAUDE CODE

USAGE PATTERNS

- Peak activity at 16:00–17:00
- Consistent usage during working hours (09:00–17:00)
- Indicates strongest engagement toward end-of-day workflows

SESSION BEHAVIOR

- Average 4–5 prompts per session
- Short, iterative interactions rather than long conversations

TOOL USAGE

- Most frequent tools:
- Read (code exploration)
- Bash (execution)
- Edit (modification)
- Reflects standard development loop:
- Read → Modify → Execute

EFFICIENCY OBSERVATIONS

- ~70% of tool actions auto-approved (low friction)
- Bash has lowest success rate (~93%)
- → Indicates execution complexity and environment issues

COST & MODEL ANALYSIS

Overall Cost

- ~$6,000 TOTAL SPEND OVER 60 DAYS
- ~100 ENGINEERS USING SYSTEM

Model Distribution

- OPUS MODELS
- ~70% OF TOTAL COST
- ~40% OF REQUESTS
- → HIGH CAPABILITY BUT EXPENSIVE
- HAIKU MODEL
- ~39% OF REQUESTS
- MINIMAL COST CONTRIBUTION
- → HIGHLY EFFICIENT FOR SIMPLE TASKS

Organizational Insights

- HIGHEST SPENDING TEAMS:
- ML ENGINEERING
- FRONTEND ENGINEERING
- HIGHEST USAGE GROUP:
- MID-LEVEL ENGINEERS (L5–L6)

PREDICTIVE ANALYTICS & RECOMMENDATIONS

FORECASTING

- Linear Regression predicts stable usage
- Estimated cost: ~$100 per day (next 14 days)
- No exponential growth → controlled adoption

ANOMALY DETECTION

- 2σ threshold identifies 3–4 abnormal high-cost days
- Likely caused by intensive workloads or batch operations

COST DRIVERS

Random Forest model shows:

- Request count = strongest predictor of cost
- Followed by seniority level and practice

USAGE PATTERNS

- Isolation Forest detects spikes at 16–17h
- Confirms behavioral trend from earlier analysis

THANKYOU